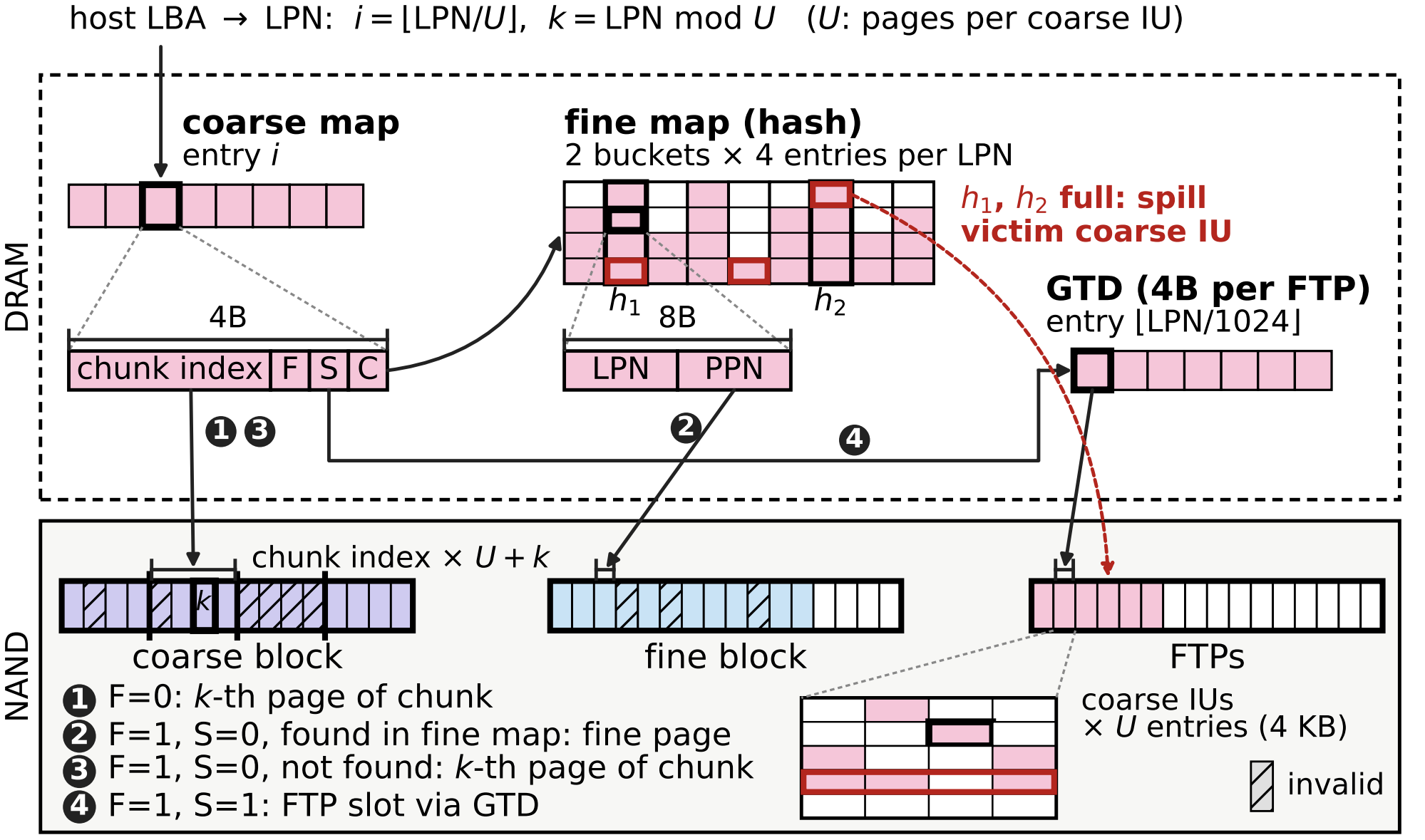
host LBA → LPN: $i = \lfloor LPN/U \rfloor$, $k = LPN \bmod U$ (U: pages per coarse IU)
DRAM
coarse map
entry i
4B
chunk index
F
S
C
fine map (hash)
2 buckets × 4 entries per LPN
h_1
h_2
8B
LPN
PPN
h_1, h_2 full: spill
victim coarse IU
GTD (4B per FTP)
entry ⌊LPN/1024⌋
1 3
2
4
NAND
chunk index × U + k
k
coarse block
fine block
FTPs
coarse IUs
× U entries (4 KB)
invalid
1 F=0: k-th page of chunk
2 F=1, S=0, found in fine map: fine page
3 F=1, S=0, not found: k-th page of chunk
4 F=1, S=1: FTP slot via GTD